

Timothy Southwick

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Senior Firmware Engineer with 10+ years developing embedded systems, Linux drivers, and FPGA development.

EDUCATION

University of Massachusetts Lowell, Lowell, MA

Master of Engineering in Computer Engineering, GPA: 3.5/4.0

Focus: Digital processor design: hardware and drivers

Bachelor of Engineering in Computer Engineering, GPA: 3.1/4.0

RELEVANT SKILLS

Programming Languages: Java, C/C++, Rust, Python 3, Python 2, Verilog, SystemVerilog, JSON, YAML, Lua, BASH, BASIC, Assembly

Skills: Digital processor design, embedded systems, Linux kernel modules, Linux drivers, real-time processing, feature switches, cache design, asynchronous and multithreaded design, data structure optimization, reading IC data sheets, computer graphics

Software: Linux, Git, GitHub, SSH, Gradle, Maven, Redis, RabbitMQ, Kubernetes, Docker, screen, tmux, AI, ALSA

Hardware: FPGA, I2C, PIC Microcontrollers, Raspberry Pi, Arduino, Lego robotics (using C)

EXPERIENCE

Lead Developer for Monumenta Games LLC, Remote (2017-Present)

- Lead distributed development across 50+ global contributors, coordinating architecture decisions and code reviews.
- Deploy and maintain 130+ Kubernetes Docker deployments on several domains across multiple dedicated servers, implementing RabbitMQ messaging and Redis data storage.
- Spearheading cross-server Java plugins, including:
 - Chat plugin with configuration saved as JSON in Redis and messages via RabbitMQ.
 - Developed an item mail system with caching and duplication protection algorithms.
 - Rewrote and ported to a guild system for better scaling and fine-grained permissions.
- Leading Python/Rust automation programs for weekly updates and maintenance.
- Implemented a custom parser of JSON-like data for use in a data editing tool.
- Devised a space partitioning data structure based on k-d trees to identify which zone, if any, a point in 3D space occupies in $O(\log(n))$ time.

Firmware Engineer for Doble Engineering, Watertown, MA (July 2019-July 2021)

- Developed high-level embedded firmware for signal generation, data acquisition, and signal analysis for use in power transmission and protection equipment testing.
- Made use of the Scrum agile collaboration framework
- Improved file transfer and debugging a DHCP network issue.
- Automated collection and processing of error code logs for quarterly reports.
- Implemented memory-efficient background data transfer process, optimizing for constrained

embedded environments.

- Firmware in C++ via gcc, debugging with gdb, testing with Python, and committed through Git.

Personal Projects

Full-time Linux user

(2009-Present)

- Home desktop doubles as an SSH/SFTP file server with public key authentication.
- Administrative tasks such as installing and maintaining server software (boot loader, sshd, MySQL, JDK, Kubernetes).
- Created automated installation scripts to install programs from the package manager and share personal files between multiple operating systems with separate program configurations.

Firmware programmer

(2005-Present)

- Built and programmed robots in C with analog/digital sensors integration and PWM/PID motor control, including a 24 hour time-lapse panorama system with automated camera rotation and image stitching.
- Used an Arduino to automate climate control at home using smart outlets.
- Developed device driver and initialization code for hardware projects, interpreting IC datasheets and hardware specifications.
- Studied the original USB Type-C specifications.

Application level programmer

(2009-Present)

- Automated phone routines using BASIC, such as using my calendar to silence it and run alarms.
- Scripted the archival of old pictures by year and month at full quality.
- Extensive use of the AI course from UC Berkeley for problem solving.
- Wrote 150+ programs for daily use and amusement in BASH, Python, Lua, C, and assembly.

Capstone Project

(2016)

- Developed audio device drivers per the ALSA framework and I²C communication protocols.
- Audio Processor for Mobile Audio Editing and Recording.
- Used the BeagleBone Black, Debian, C, I²C, ALSA, and SSH to record, edit, and play audio.

SystemVerilog

(2016-2018)

- Developed a SIMD Connect Four engine with custom VGA display controller in SystemVerilog for an FPGA implementation.
- Performed verification exercises on sample code.